THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING MECHANICS AND MATERIALS – CIVL1013

DATE: 5 May, 2009 TIME: 9:30 am – 12:30 pm (3 hours)

Answer any 2 questions from Section A and any 2 questions from Section B.
All questions carry equal marks.
Use a separate answer book for Section A and Section B.

Use of Electronic Calculators:

Candidates may use any calculator which fulfils the following criteria:

(a) it should be self-contained, silent, battery-operated and pocket-sized;
(b) it should have numeral-display facilities only and should be used only for the purpose of calculation;
(c) it should not have any printing device, alphanumeric keyboard, or graphic display; and
(d) it should not contain any recorded data or program.

It is the candidate's responsibility to ensure that the calculator operates satisfactorily and the candidate must record the name and type of the calculator on the front page of the examination scripts. Lists of permitted/prohibited calculators will not be made available to candidates for reference, and the onus will be on the candidate to ensure that the calculator used will not be in violation of the criteria listed above.

# SECTION A

You are required to answer 2 out of 3 questions in Section A.

**Question 1**

Consider the simply supported beam given in Figure 1 in which E = 200,000 MPa and $I = 10.0 \times 10^{-4}$ m$^4$.

(a) Calculate all reactions (include suitable check). (**6 marks**)
(b) Derive the bending moment equation for the entire beam by using Macaulay's bracket functions. (**5 marks**)
(c) Calculate the deflection at point C by utilising the result obtained in part 1b (**10 marks**).
(d) Describe the First Moment-Area Theorem and provide suitable diagram/s where appropriate. (**4 marks**)

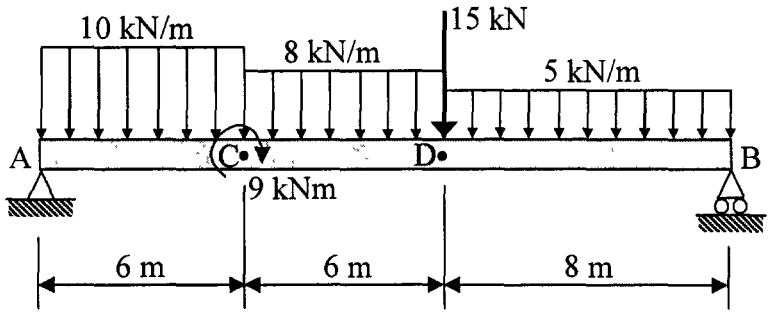



Figure 1

**Question 2**

Consider the rectangular simply supported beam with overhang, of depth 200 mm and width 150 mm, as given in Figure 2.

(a) Calculate all reactions (include suitable check). (**6 marks**)
(b) Calculate the internal stress resultants of shear and bending moment at positions (i) just to the left of reaction B, and (ii) just to the right of reaction B. (**6 marks**)
(c) Draw the shear force diagram for the entire beam. (**6 marks**)
(d) Draw the tension side bending moment diagram for the entire beam. (**7 marks**)

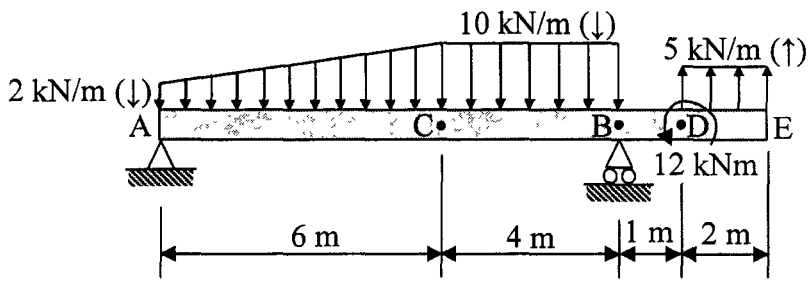



Figure 2

**Question 3**

(a) Consider the plane stress element given in Figure 3. Construct Mohr's circle and then calculate the principal stresses (you do not need to calculate the orientation of the principal planes). (**9 marks**)

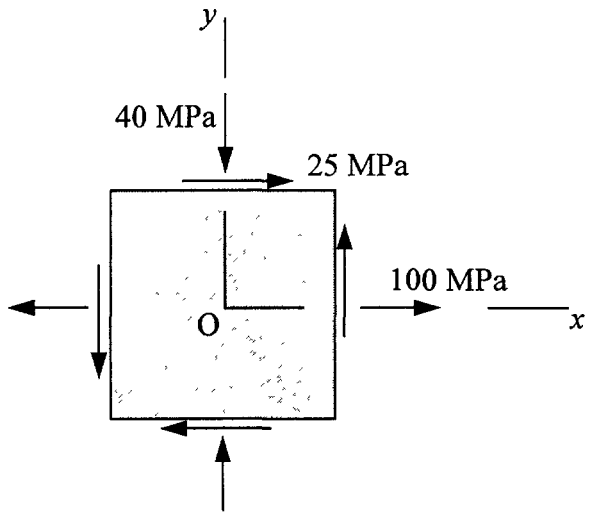



Figure 3

(b) A pinned-end mild-steel strut (E = 200,000 MPa) of length 5 m is made of a rectangular square hollow section of sectional depth 150 mm and width 75 mm (Figure 4). Calculate the wall thickness t required to produce a buckling capacity of 50 kN. (**10 marks**)

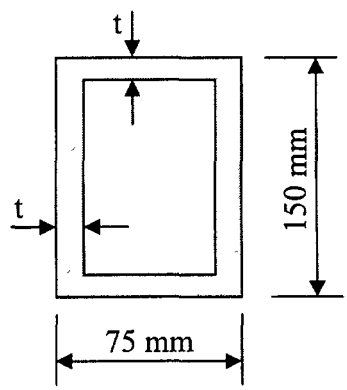



Figure 4

(c) Plot the following relationship for structural steel on a graph of x-axis representing slenderness and y-axis representing stress. Show all necessary information on your plot and assume $\sigma_y$ = 300 MPa and E = 200,000 MPa. (**6 marks**)

$$\sigma_{cr} = \frac{\pi^2 E}{\left(L_e/r\right)^2} \leq \sigma_y$$

**- END OF SECTION A -**